MintpathOS

The First Biologic Autonomous Kernel

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Production Validated · Revenue Generating

Patent Pending

Classification: Level 3 Autonomous Infrastructure

This document describes system behavior and architecture. Certain mechanisms are intentionally abstracted.

1. Executive Summary

MintpathOS is the first production-validated Level 3 Autonomous Kernel capable of planning, executing, evaluating, and safely evolving its own production codebase under strict governance, legality, and audit constraints. Level 3 autonomy denotes systems that independently operate and adapt without human-in-the-loop approval for routine evolution, while remaining bounded by non-overridable policy constraints.

Built by a single operator without formal software training, MintpathOS compresses traditional engineering timelines by orders of magnitude. The system has grown from initial concept to 190,000+ lines of production TypeScript across 1,000+ files and 50+ subsystems in approximately four months. The majority of this was produced in a single intensive sprint.

MintpathOS is live in production, generates revenue through autonomous vulnerability discovery, and includes CLI, web, and mobile PWA interfaces for governed operational control. It routes across four major LLM providers and 50+ models via policy-driven selection.

2. System Architecture Overview

MintpathOS is structured as a multi-layered autonomous organism designed to maintain coherence, safety, and deterministic evolution. Eight independent worker processes run continuously under process supervision.

2.1 SQL Skeleton (Structural Truth)

The SQL layer forms the immutable backbone of system reality through an Auto-Safe SQL Evolution Engine enforcing additive-only evolution, atomic migrations, rollback isolation, and hard invariants across 30+ tables.

2.2 Semantic Control Plane (Cognition)

The TypeScript control plane acts as a bilingual reasoning layer bridging rigid SQL structure with dynamic logical meaning, coordinating job pipelines and worker nodes.

Cerebrum is the long-lived authoritative reality substrate, exposing queryable faculties (RepoFaculty, DbFaculty, FsFaculty) to ensure fresh, tool-backed reads without serialized snapshots or drift.

2.3 Liquid Crystal Architecture (Schema Immune System)

Liquid Crystal enforces coherence across SQL schemas, TypeScript types, JSON contracts, and runtime validators to prevent drift and silent failures.

2.4 Architect Router (Self-Healing Dispatch)

The Architect Router autonomously detects malformed or stalled intentions and normalizes them into executable envelopes.

Padlock Substrate provides a drift-resistant execution layer with hash-locked JobSpecs, restricted IR opcodes, mandatory verifiers, and fail-closed invariants.

PatchPlan defines the canonical mutation primitive: idempotent, multi-file plans with runtime validation and quarantine-backed execution.

3. Recursive Coherence

MintpathOS exhibits Recursive Coherence—the capacity to recognize and repair its own schemas, pipelines, and reward mappings. This is demonstrated in production via full self-evolution cycles: intent → governed patch → root build → auto-commit/push.

The Code Repair Framework (CRF) and MintPatch engine provide snapshot-based verified code mutation with cryptographic rollback, enabling the system to safely modify its own source under governance constraints.

4. Governor: Cryptographic Execution Authority

Governor is the sole authority for state mutation, enforcing signed envelopes, deterministic payloads, and regulator-grade auditability with immutable logs. All agent actions are routed through policy enforcement with risk budgets. Agents exceeding their risk allocation are quarantined automatically.

Role-based access control (RBAC) defines permission boundaries. Mutation contracts validate all state changes before execution.

5. Epoch & Validity Kernels: Cryptographic Trust Anchoring

The Epoch system provides immutable execution records and evidence-based claim aggregation. Every autonomous action produces a sealed artifact in an intermediate representation (IR), which is evaluated by Validity Kernels (VKs) for integrity.

Claims are fused through weighted evidence aggregation and must pass promotion gates before elevation. This creates a provenance chain where every decision the system makes is cryptographically attributable and replayable.

6. Multi-Agent Distributed Dispatch (MADD)

MADD decomposes complex tasks into skill-based execution waves. Tasks are first-class entities with their own governance policies, not entries in a job queue.

Parallel wave execution enables concurrent skill application. Automatic healing detects and repairs failed task branches without human intervention. A dedicated skill registry maps capabilities to available agents.

7. Liquid Control Plane & Live State Gateway

The Liquid Control Plane (LCP) provides a unified control surface over all system projections through five primitives: the World Graph (structural entity map), Event Spine (ordered event stream), Control Intents (high-level action requests), Guarded State (safety-enforced mutations), and Change Sheets (immutable mutation artifacts with rollback).

The Live State Gateway (LSG) implements real-time bidirectional state synchronization via SSE and subscriptions, with automatic reconciliation and resilience wrappers. Browser and backend remain in sync without polling.

8. M-CON v2: Intent to Operation Compiler

M-CON v2 replaces free-form prompting with deterministic compilation of intent into bounded, typed operations, allowing natural language job execution without brittle JSON prompts.

9. Biologic Affective Control Substrate

MintpathOS maintains a deterministic affective state modulating execution posture using arousal, valence, trust, and workload metrics grounded in first-party datasets. These are behavioral substrates—hardwired state machines governing agent behavior—not LLM-delegated personality traits.

The persona biologic, kernel biologic, and transition biologic operate as layered behavioral constraints that shape how agents communicate, prioritize, and escalate.

10. Memory & Situational Awareness

The Memory system provides persistent recall via vector embeddings with semantic and deterministic retrieval. A prompt assembler injects relevant memory into LLM context dynamically.

Peephole provides real-time situational awareness through file system and database watchers. Changes to code, configuration, or data trigger LLM-orchestrated responses via a playbook system. The system observes its own environment continuously.

11. Multi-Model Intelligence Routing

MintpathOS routes across four LLM providers and 50+ models via a policy-driven model registry:

- Anthropic (Claude Opus, Sonnet, Haiku)
- OpenAI (GPT-5.x, GPT-4o, o3/o1 reasoning series, Codex)
- Google Vertex AI (Gemini 3.x, 2.5 Pro/Flash)
- XAI (Grok 4.x series)

Model selection is capability-driven: reasoning tasks route to reasoning models, code generation to code-specialized models, fast classification to lightweight models. Batch processing APIs reduce cost for non-latency-sensitive workloads. Provider failover is automatic.

12. Autonomous Revenue Generation

MintRecon is a fully integrated vulnerability discovery subsystem that autonomously scans bug bounty targets across Immunefi, HackerOne, HackenProof, and Cantina. It operates as a continuous 5-minute cycle worker process.

Findings are emitted as epoch evidence, scored by yield metrics, and routed through the governance system. The subsystem adapts its scanning strategy based on historical probe performance.

MintRecon has produced actionable Critical and High severity findings against production targets within its first weeks of operation.

13. LUX: Governed UI Transform Kernel

LUX governs visual transformations via registered ops, visual anchors, and risk-based dampening. The MintOps PWA and Next.js web interface provide secure access to governed operations across desktop and mobile.

14. Grounded Model Pipeline

All model inference and learning rely on first-party logs, deterministic snapshots, and governed lineage to structurally bound hallucinations. No external training data enters the system without provenance tracking.

15. Conclusion

MintpathOS establishes a new paradigm for autonomous infrastructure: biologic control with structural rigor, preserving human sovereignty while enabling safe autonomous evolution.

The system demonstrates that a governed autonomous kernel—built by a single non-engineer operator using AI as the primary development instrument—can reach production scale, generate revenue, and maintain the safety properties required for trusted autonomous operation.

Metric	**Value**
Lines of Code	190,000+
Source Files	1,000+
Major Subsystems	50+
Worker Processes	8
LLM Models Supported	50+
LLM Providers	4
Database Tables	30+
Development Time	∼4 months
Developer Count	1 (non-engineer)

Appendix A: Deterministic Replayability

Every autonomous action is replayable via cryptographic snapshots, pure kernel functions, and immutable ledgers. The Epoch/VK system ensures that execution paths are sealed at the point of decision, enabling full forensic reconstruction of any autonomous action the system has taken.

Appendix B: Architectural Primitives

The following primitives constitute the novel IP surface of MintpathOS. Each operates as an independent, licensable substrate:

- **Governor / Autonomy Bus** — Policy-based autonomy control with risk budgets and deterministic routing
- **Epoch / Validity Kernels** — Cryptographic trust anchoring for AI execution paths
- **Biologic Kernels** — Behavioral substrates driving agent decisions contextually
- **Peephole** — Real-time filesystem and database watchers for situational awareness
- **CRF / MintPatch** — Snapshot-based verified code mutation with rollback
- **MADD** — Skill-based multi-agent task decomposition with wave parallelism
- **Liquid Control Plane** — Unified control surface with World Graph, Event Spine, and Guarded State
- **Live State Gateway** — Real-time bidirectional state synchronization
- **Memory System** — Persistent recall via embeddings with semantic retrieval
- **M-CON v2** — Deterministic intent-to-operation compilation
- **Liquid Crystal** — Cross-layer schema coherence enforcement
- **MintRecon** — Autonomous vulnerability discovery and revenue generation